STAT6011

COMPUTATIONAL STATISTICS AND BAYESIAN LEARNING

Lecture 1

Bayesian Methods and Sampling Techniques

Bayesian Inference

- The philosophical distinction between Bayes and frequentists is deep.
- In the frequentist framework, a probability is representing some type of long run frequency, i.e. when we say the probability that our estimator is close to some unknown "true" parameter with probability at least $1 - \delta$ we are really imagining repeating this (or some other) experiment many many times and then our guarantees will be correct for at least $1 - \delta$ of these experiments.
- Similarly, with confidence intervals, we imagine many people across the world construct confidence intervals and our guarantee is that 95% of those intervals would trap the true parameter.
- The goal of frequentist inference is to create procedures with long run guarantees.

- The guarantees should be uniform over θ if possible. For example, a confidence interval traps the true value of θ with probability $1 - \alpha$, no matter what the true value of θ is.
- In frequentist inference, procedures are random while parameters are fixed, unknown quantities.
- In the Bayesian approach, probability is regarded as a measure of subjective degree of belief.

Bayesian Statistics

- Bayes' rule: if A and B are two events in the sample space Ω, then the conditional probability of A given B is

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(B|A)P(A)}{P(B)},$$

  where $P(B) \neq 0$.

- If $P(A \cap B) = P(A)P(B)$, A and B are independent.

- More generally, if $\{A_1, \ldots, A_m\}$ is a partition of Ω, then for each $i = 1, \ldots, m$,

$$P(A_i|B) = \frac{P(B|A_i)P(A_i)}{\sum_{j=1}^{m} P(B|A_j)P(A_j)}.$$

Likelihood

- Let $\mathbf{y} = \{y_1, \ldots, y_n\}$ be an i.i.d. sample from a density function $f(y|\boldsymbol{\theta})$ which is characterized by a vector of unknown parameters $\boldsymbol{\theta}$.
- The likelihood function is

$$L(\boldsymbol{\theta}|\mathbf{y}) \equiv p(\mathbf{y}|\boldsymbol{\theta}) = \prod_{i=1}^{n} f(y_i|\boldsymbol{\theta}).$$

- The maximum likelihood estimator (MLE) is obtained by maximizing $L(\boldsymbol{\theta}|\mathbf{y})$ or $\log L(\boldsymbol{\theta}|\mathbf{y})$.
- The data are random and the parameters are fixed in the frequentist perspective.

- The data are observed (fixed) and the parameters are random in the Bayesian paradigm.
- Given a prior distribution $p(\boldsymbol{\theta})$, the posterior distribution of $\boldsymbol{\theta}$ is given by

$$p(\boldsymbol{\theta}|\mathbf{y}) \propto p(\mathbf{y}|\boldsymbol{\theta})p(\boldsymbol{\theta}).$$

- We may incorporate the normalizing constant to make $p(\boldsymbol{\theta}|\mathbf{y})$ a valid density function; that is,

$$p(\boldsymbol{\theta}|\mathbf{y}) = \frac{p(\mathbf{y}|\boldsymbol{\theta})p(\boldsymbol{\theta})}{\int p(\mathbf{y}|\boldsymbol{\theta})p(\boldsymbol{\theta})\,\mathrm{d}\boldsymbol{\theta}}.$$

Posterior Predictive

- As one step further, the Bayesian posterior predictive distribution is concerned with a future observation $\tilde{y}$ conditional on the current data $\mathbf{y}$.
- If $\tilde{y}$ and $\mathbf{y}$ are conditionally independent given $\boldsymbol{\theta}$, which is often the case, then

$$f(\tilde{y}, \boldsymbol{\theta}|\mathbf{y}) = f(\tilde{y}|\boldsymbol{\theta}, \mathbf{y})p(\boldsymbol{\theta}|\mathbf{y}) = f(\tilde{y}|\boldsymbol{\theta})p(\boldsymbol{\theta}|\mathbf{y}).$$

- By integrating out $\boldsymbol{\theta}$ from $f(\tilde{y}, \boldsymbol{\theta}|\mathbf{y})$, we obtain the posterior predictive distribution,

$$f(\tilde{y}|\mathbf{y}) = \int f(\tilde{y}|\boldsymbol{\theta})p(\boldsymbol{\theta}|\mathbf{y})\, \mathrm{d}\boldsymbol{\theta}.$$

Normal Example

- Let $\mathbf{y} = \{y_1, \ldots, y_n\}$ be an i.i.d. sample from the normal distribution $\mathrm{N}(\mu, \sigma^2)$. For simplicity, suppose that the variance σ^2 is known.
- We are interested in estimating μ using the Bayesian approach. First, we specify a prior distribution, $\mu \sim N(\mu_0, \sigma_0^2)$.
- Given the observed data $\mathbf{y}$, the likelihood function is

$$L(\mu|\mathbf{y}) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi}\sigma} \exp\left\{ -\frac{(y_i - \mu)^2}{2\sigma^2} \right\}.$$

- Then, the posterior distribution of μ is given by

$$\begin{aligned} p(\mu|\mathbf{y}) &\propto \exp\left\{ -\frac{\sum_{i=1}^{n}(y_i-\mu)^2}{2\sigma^2} \right\} \times \exp\left\{ -\frac{(\mu-\mu_0)^2}{2\sigma_0^2} \right\} \\ &\propto \exp\left\{ -\frac{1}{2}\left(\frac{\sigma_0^2+\sigma_n^2}{\sigma_0^2\sigma_n^2}\right)\left(\mu - \frac{\sigma_0^2\bar{y}_n+\mu_0\sigma_n^2}{\sigma_0^2+\sigma_n^2}\right)^2 \right\}, \end{aligned}$$

  where $\bar{y}_n$ is the sample mean and $\sigma_n^2 = \sigma^2/n$ is the corresponding variance.

- Clearly, the posterior distribution of μ is still normal with mean $(\sigma_0^2\bar{y}_n + \mu_0\sigma_n^2)/(\sigma_0^2+\sigma_n^2)$ and variance $\sigma_0^2\sigma_n^2/(\sigma_0^2+\sigma_n^2)$.

- The posterior mean of μ can be viewed as a weighted average of the prior mean μ_0 and the sample mean $\bar{y}_n$, weights inversely proportional to their variances.

Binomial Distribution

- $Y \sim \text{Bin}(n, p)$, if $Y = X_1 + \cdots + X_n$, where $X_1, \ldots, X_n$ are independent and identically distributed (i.i.d.) random variables from $\text{Bernoulli}(p)$.
- The p.m.f. of Y is given by

$$\begin{aligned} P(Y = y) &= \binom{n}{y} p^y (1-p)^{n-y} \\ &= \frac{n!}{y!(n-y)!} p^y (1-p)^{n-y}, \quad y = 0, 1, \ldots, n, \end{aligned}$$

with $\mathsf{E}(Y) = np$ and $\mathsf{Var}(Y) = np(1-p)$.

Beta Distribution

- $X \sim \text{Beta}(\alpha, \beta)$, if the p.d.f. of X is

$$f(x) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1}(1-x)^{\beta-1}, \quad 0 < x < 1,$$

where the two shape parameters $\alpha > 0$ and $\beta > 0$, and

$$\mathsf{E}(X) = \frac{\alpha}{\alpha+\beta} \quad \text{and} \quad \mathsf{Var}(X) = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}.$$

- When $\alpha = \beta = 1$, the beta distribution reduces to the uniform distribution on $(0, 1)$.
- The beta distribution is typically used as a prior distribution for a probability parameter.

Conjugate Prior Distribution

- Let θ be the response rate of an experimental drug, and let $p(\theta)$ be the prior distribution of θ, which represents uncertainty on θ before observing any data.
- We take $\theta \sim \mathrm{Beta}(\alpha, \beta)$; that is, a beta prior distribution for θ with the density function of

$$p(\theta) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}\theta^{\alpha-1}(1-\theta)^{\beta-1}.$$

- The parameters α and β in the beta prior distribution can be viewed as the numbers of prior successes and failures, respectively; $\alpha + \beta$ can be regarded as the number of prior observations measuring how informative the prior distribution is.
- For example, $\mathrm{Beta}(0.5, 1.5)$ is a vague prior distribution, which contains information as much as two observations only, while $\mathrm{Beta}(5, 15)$ is a more informative prior distribution, which contains the amount of information corresponding to 20 subjects.

Likelihood and Posterior

- If we observe y responders among n subjects treated by the investigational drug, the likelihood under the binomial distribution is

$$p(y|\theta) = \binom{n}{y}\theta^y(1-\theta)^{n-y}.$$

- The posterior distribution for the response rate θ is given by

$$p(\theta|y) \propto p(y|\theta)p(\theta) \propto \theta^{\alpha+y-1}(1-\theta)^{\beta+n-y-1}.$$

- The posterior distribution of θ is also a beta distribution,

$$\theta|y \sim \text{Beta}(\alpha+y, \beta+n-y). \tag{1}$$

Example

- As an illustration, we specify a beta prior distribution for the response rate of an experimental drug, $\theta \sim \mathrm{Beta}(2, 6)$.
- If we observe $y = 15$ responders among $n = 25$ patients treated in the trial, the posterior distribution of θ is updated as $\theta|y \sim \mathrm{Beta}(17, 16)$.
- Figure 1 shows that from the prior to the posterior distribution of θ, the distribution is shifted to the right in light of the observed data.

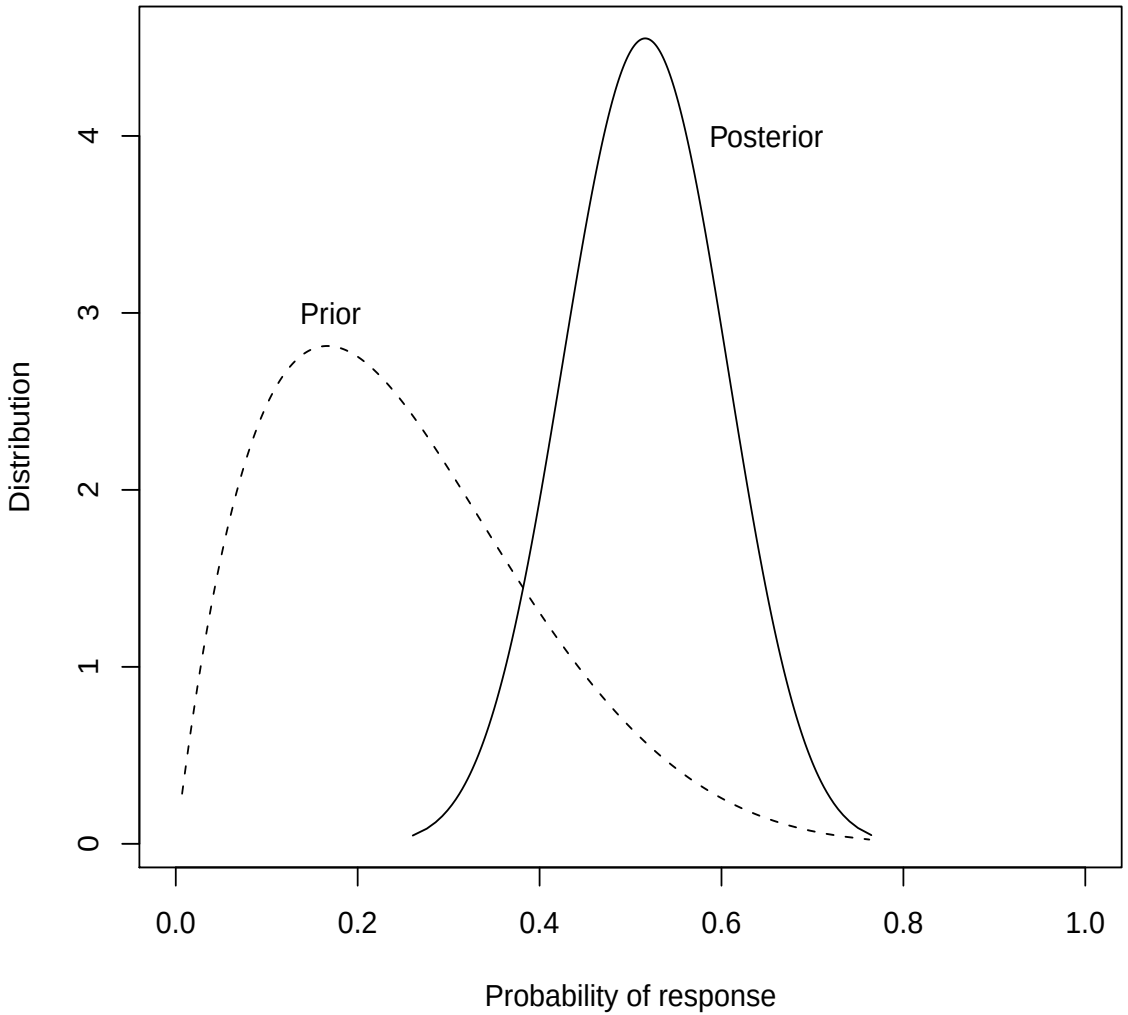


Figure 1: Update the probability of response from the prior to the posterior distribution using a beta prior distribution.

Multinomial Distribution

- For $i = 1, \ldots, K$, let p_i denote the probability of the occurrence of event i, $\sum_{i=1}^{K} p_i = 1$; and let x_i denote the number of occurrences of event i in a total of n experiments, $\sum_{i=1}^{K} x_i = n$.
- The random vector **X** follows a multinomial distribution, if the joint p.m.f. of **X** is

$$f(x_1, \ldots, x_K) = \frac{n!}{x_1! \cdots x_K!} p_1^{x_1} \cdots p_K^{x_K}.$$

- Each X_i marginally follows a binomial distribution, $X_i \sim \text{Bin}(n, p_i)$; and $\text{E}(X_i) = np_i$, $\text{Var}(X_i) = np_i(1 - p_i)$, and $\text{Cov}(X_i, X_j) = -np_ip_j$ for $i \neq j$.

Dirichlet Distribution

- $\mathbf{X} \sim \mathrm{Dir}(\alpha_1, \ldots, \alpha_K)$, if $\mathbf{X}$ has a p.d.f. of

$$f(\mathbf{x}) = \frac{\Gamma(\alpha)}{\Gamma(\alpha_1)\cdots\Gamma(\alpha_K)} x_1^{\alpha_1 - 1} \cdots x_K^{\alpha_K - 1},$$

  where $\alpha = \sum_{i=1}^K \alpha_i$ with $\alpha_i > 0$, and $\sum_{i=1}^K x_i = 1$ with $0 < x_i < 1$.

- Each X_i marginally follows a beta distribution, $X_i \sim \mathrm{Beta}(\alpha_i, \alpha - \alpha_i)$, with $\mathrm{E}(X_i) = \alpha_i/\alpha$,

$$\mathrm{Var}(X_i) = \frac{\alpha_i(\alpha - \alpha_i)}{\alpha^2(\alpha + 1)} \quad \text{and} \quad \mathrm{Cov}(X_i, X_j) = -\frac{\alpha_i \alpha_j}{\alpha^2(\alpha + 1)} \ (i \neq j).$$

- The Dirichlet distribution is a multivariate version of the beta distribution, which reduces to a beta distribution when $K = 2$.
- In addition, for any $i \neq j$,

$$(X_1, \ldots, X_i + X_j, \ldots, X_K)^\top \sim \mathrm{Dir}(\alpha_1, \ldots, \alpha_i + \alpha_j, \ldots, \alpha_K).$$

Multinomial and Dirichlet

- The multinomial distribution is a generalization of the binomial distribution, which is particularly useful for monitoring multivariate outcomes in a clinical trial.
- Correspondingly, the Dirichlet distribution is the multivariate version of the beta distribution.
- For $k = 1, \ldots, K$, let θ_k denote the probability of the occurrence of the kth event, satisfying $\sum_{k=1}^{K} \theta_k = 1$.
- Define $\boldsymbol{\theta} = (\theta_1, \ldots, \theta_K)^\top$; then we can specify a Dirichlet prior distribution for $\boldsymbol{\theta}$,

$$p(\boldsymbol{\theta}) = \frac{\Gamma(\alpha_1 + \cdots + \alpha_K)}{\Gamma(\alpha_1) \cdots \Gamma(\alpha_K)} \theta_1^{\alpha_1 - 1} \cdots \theta_K^{\alpha_K - 1}.$$

Likelihood and Posterior

- If the observed data $\mathbf{y} = \{y_1, \ldots, y_K\}$ follow a multinomial distribution with $\sum_{k=1}^{K} y_k = n$, the likelihood function is

$$L(\boldsymbol{\theta}|\mathbf{y}) = \frac{n!}{y_1! \cdots y_K!} \theta_1^{y_1} \cdots \theta_K^{y_K}.$$

- Given the prior distribution $\boldsymbol{\theta} \sim \mathrm{Dir}(\alpha_1, \ldots, \alpha_K)$, the posterior distribution of $\boldsymbol{\theta}$ is

$$p(\boldsymbol{\theta}|\mathbf{y}) \propto \theta_1^{\alpha_1 + y_1 - 1} \cdots \theta_K^{\alpha_K + y_K - 1}.$$

- That is also a Dirichlet distribution due to the conjugate property,

$$\boldsymbol{\theta}|\mathbf{y} \sim \mathrm{Dir}(\alpha_1 + y_1, \ldots, \alpha_K + y_K),$$

Example: Posterior Expectations

- Posterior distribution

$$\begin{cases} f(\boldsymbol{y}|\theta) = \prod_{i=1}^{n} f(y_i|\theta) \\ \pi(\theta) = \text{prior on } \theta \end{cases} \Rightarrow f(\theta|\boldsymbol{y}) = \frac{f(\boldsymbol{y}|\theta)\pi(\theta)}{\int f(\boldsymbol{y}|\theta)\pi(\theta)d\theta}$$

- Flat prior $\pi(\theta) \propto 1$.
- What is the estimator: posterior mean, MAP? Maximum a posteriori probability (MAP) estimate equals the mode of the posterior distribution.

Analytic Approximation

- The Laplace approximation is an extraordinarily useful technique for approximating expectations.
- This approximation replaces the kernel of an integrand with a Taylor series centered at the mode.
- Solve $\int_a^b f(x;\theta)dx$; if $f(x,\theta)$ becomes increasingly centered about its mode $\hat{x}_\theta$, then the Laplace approximation is asymptotically valid.
- The approximation improves as $|b-a|$ becomes smaller or $a, b \approx \hat{x}_\theta$.

- Taylor's expansion around the mode $\hat{x}_\theta$,

$$
\begin{aligned}
& P(X \in [a,b]) = F_\theta(b) - F_\theta(a) \\
= & \int_a^b f(x;\theta)dx \\
= & \int_a^b \exp\{h(x;\theta)\}dx \quad (\text{Let } h(x;\theta) = \log f(x;\theta)) \\
\approx & \int_a^b \exp\{h(\hat{x}_\theta;\theta) + h'(\hat{x}_\theta;\theta)(x-\hat{x}_\theta) + \frac{h''(\hat{x}_\theta;\theta)}{2}(x-\hat{x}_\theta)^2\}dx \\
= & \exp\{h(\hat{x}_\theta;\theta)\} \int_a^b \underbrace{\exp\left\{\frac{-(x-\hat{x}_\theta)^2}{2(-h''(\hat{x}_\theta;\theta))^{-1}}\right\}}_{\text{normal kernel}} dx \\
= & \exp\{h(\hat{x}_\theta;\theta)\}\left(\frac{2\pi}{-h''(\hat{x}_\theta;\theta)}\right)^{\frac{1}{2}} \\
& \times\left[\Phi\left(\frac{b-\hat{x}_\theta}{(-h''(\hat{x}_\theta;\theta))^{-\frac{1}{2}}}\right) - \Phi\left(\frac{a-\hat{x}_\theta}{(-h''(\hat{x}_\theta;\theta))^{-\frac{1}{2}}}\right)\right]
\end{aligned}
$$

Maximum a Posteriori estimation: MAP

- The Laplace approximation to the posterior mean is the posterior mode.

$$
\begin{aligned}
& \text{posterior mean} \\
= \quad & \frac{\int \theta f(\boldsymbol{y}|\theta)\pi(\theta)d\theta}{\int f(\boldsymbol{y}|\theta)\pi(\theta)d\theta} = \frac{\int \theta \exp\{h(\theta|y)\}d\theta}{\int \exp\{h(\theta|y)\}d\theta} \\
\approx \quad & \frac{\int \theta \exp\{h(\hat{\theta}|y) + h''(\hat{\theta}|y)(\theta-\hat{\theta})^2/2\}d\theta}{\int \exp\{h(\hat{\theta}|y) + h''(\hat{\theta}|y)(\theta-\hat{\theta})^2/2\}d\theta} \\
= \quad & \frac{\int \theta \exp\{h''(\hat{\theta}|y)(\theta-\hat{\theta})^2/2\}d\theta}{\int \exp\{h''(\hat{\theta}|y)(\theta-\hat{\theta})^2/2\}d\theta} = \hat{\theta}
\end{aligned}
$$

- This is to compute the mean of a normal distribution for θ.

Monte Carlo Method

- Monte Carlo methods are a class of computational algorithms that rely on repeated random sampling to compute their results; they are used when it is infeasible or impossible to compute an exact result with deterministic algorithms.
- Let $E_f[h(x)] = \int h(x)f(x)dx$.
- If we could simulate $x_1, \ldots, x_n$ from density function $f(x)$, then we could use the Strong Law of Large Number (SLLN) to approximate

$$\sum_{i=1}^{n} h(x_i)/n \to E_f[h(x)].$$

- The estimator $\sum_{i=1}^{n} h(x_i)/n$ is approximately normally distributed about $E_f[h(x)]$ with standard error $\sqrt{\text{Var}[h(x)]/n}$.
- In practice, we estimate the standard error with $\sqrt{\upsilon/n}$, where

$$\upsilon = \frac{1}{n-1}\sum_{i=1}^{n}\left[h(x_i) - \frac{1}{n}\sum_{j=1}^{n} h(x_j)\right]^2$$

- $E_f[h(x)]$ may be a posterior mean.
- We may be interested in

$$\int h(x)dx = \int \frac{h(x)}{f(x)} f(x)dx.$$

Example: Monte Carlo Integration

- Suppose we compute $\int_0^1 h(x)dx$, where

$$h(x) = [\cos(50x) + \sin(20x)]^2.$$

- To calculate the integral, we generate $u_1, u_2, \ldots, u_n$ iid $\mathrm{Unif}(0, 1)$ random variables, and approximate $\int_0^1 h(x)dx = \int_0^1 h(x) \cdot 1dx$ with $\sum_{i=1}^n h(u_i)/n$.
- It is clear that the Monte Carlo average is converging, with a value of 0.963 after 10,000 iterations.

- This idea can be extended to compute $\int_a^b h(x)dx$, where a and b are finite numbers,

$$\int_a^b h(x)dx = (b-a)\int_a^b h(x)\frac{1}{b-a}dx \approx (b-a)\sum_{i=1}^{n} h(u_i)/n,$$

where u_i are random samples from $\mathrm{Unif}(a,b)$.

The Inversion Method

- Let X be a random variable with *cumulative distribution function* (CDF) F. Since F is a non-decreasing function, the inverse function F^{-1} may be defined by

$$F^{-1}(u) = \inf\{x : F(x) \geq u\}, \quad u \in (0, 1).$$

- $F(X) \sim \mathrm{Unif}(0, 1)$ or equivalently $X = F^{-1}(U)$ has cdf F.
- Step 1: Draw U from $\mathrm{Unif}(0, 1)$;
  Step 2: Return $X = F^{-1}(U)$.

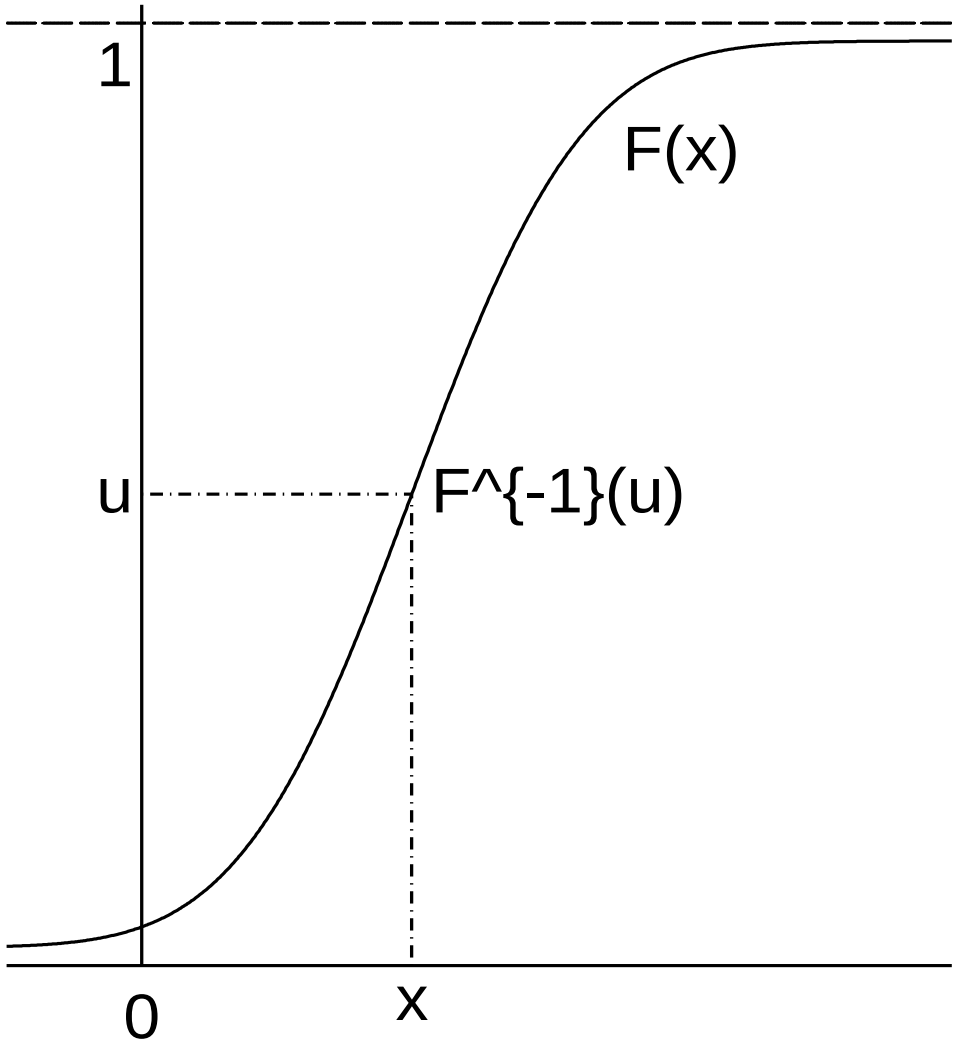


Figure 1.1 The inversion method.

Generation of Exponential Variables

- Let X follow the exponential distribution with rate parameter $\beta > 0$, then its *probability distribution function* (PDF) and CDF are given by

$$f(x) = \beta \mathrm{e}^{-\beta x}, \quad x > 0,$$

and

$$F(x) = \int_0^x f(t)\,\mathrm{d}t = 1 - \mathrm{e}^{-\beta x}, \quad x > 0,$$

respectively.

- Let $F(x) = u$, then

$$x = F^{-1}(u) = -\frac{\log(1-u)}{\beta}, \quad 0 < u < 1.$$

- If $U \sim \mathrm{Unif}(0,1)$, it is easy to verify that $1-U \sim \mathrm{Unif}(0,1)$.
- Thus, the inversion method for generating $X \sim \mathrm{Exponential}(\beta)$ is as follows:
  - ✓ Draw U from $\mathrm{Unif}(0,1)$;
  - ✓ Return $X = F^{-1}(U) = F^{-1}(1-U) = -\log(U)/\beta$. $\|$

Transformation

- If $X \sim \text{Gamma}(\alpha, 1)$, and $Y \sim \text{Gamma}(\beta, 1)$, then

$$\frac{X}{X+Y} \sim \text{Beta}(\alpha, \beta).$$

- If $\mu \sim \text{Gamma}(\alpha, \beta)$, and $Y|\mu \sim \text{Poi}(\mu)$, then $Y \sim$ Negative-Binomial.
- To simulate multivariate normals, $\boldsymbol{X} \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, let $\boldsymbol{\Sigma} = \boldsymbol{\omega}\boldsymbol{\omega}^T$ (Choleski decomposition).
- Then, $\boldsymbol{\mu} + \boldsymbol{\omega}\boldsymbol{Z}$, where $\boldsymbol{Z} \sim N(0, \boldsymbol{I})$, has the distribution $N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$.

Sequential Simulation

- We aim to simulate from $f(x, y, z) \neq f(x)f(y)f(z)$. Let $f_x(x) = \int\int f(x, y, z)dydz$,

$$f_{y|x}(y|x) = \frac{\int f(x, y, z)dz}{f_x(x)} = \frac{f_{xy}(x, y)}{f_x(x)},$$

  and $f_{z|x,y} = \frac{f(x,y,z)}{f_{xy}(x,y)}$.

- If we simulate $x_i \sim f_x(x)$, $y_i \sim f_{y|x}(y|x = x_i)$, and $z_i \sim f_{z|x,y}(z|x = x_i, y = y_i)$, then the triple (x_i, y_i, z_i) has a joint density $f(x_i, y_i, z_i)$.

- This idea extends beyond three variables, but sequential simulation can be very difficult.

Rejection Sampling

- Rejection sampling is a way to generate observations from a target density f by thinning out samples from a candidate density g.
- Notation:

f: target distribution $\quad$ g: candidate distribution

$\mathcal{X}_f$: support of f $\quad$ $\mathcal{X}_g$: support of g

- The candidate distribution function has the following properties:
  - ✓ It is relatively easy to draw random samples from $g(x)$. It is ***not*** required that $g(x)$ integrate to 1, but $g(x)$ must have a finite integral.
  - ✓ The *importance ratio* $f(x)/g(x)$ on X_f must have a known bound, i.e., $f(x)/g(x) \leq C$ for $x \in \mathcal{X}_f$, i.e.,

$$C = \sup_{x \in \mathcal{X}_f} \frac{f(x)}{g(x)} < \infty.$$

  - ✓ With $1 < C < \infty$, $Cg(x)$ is called the envelop function.

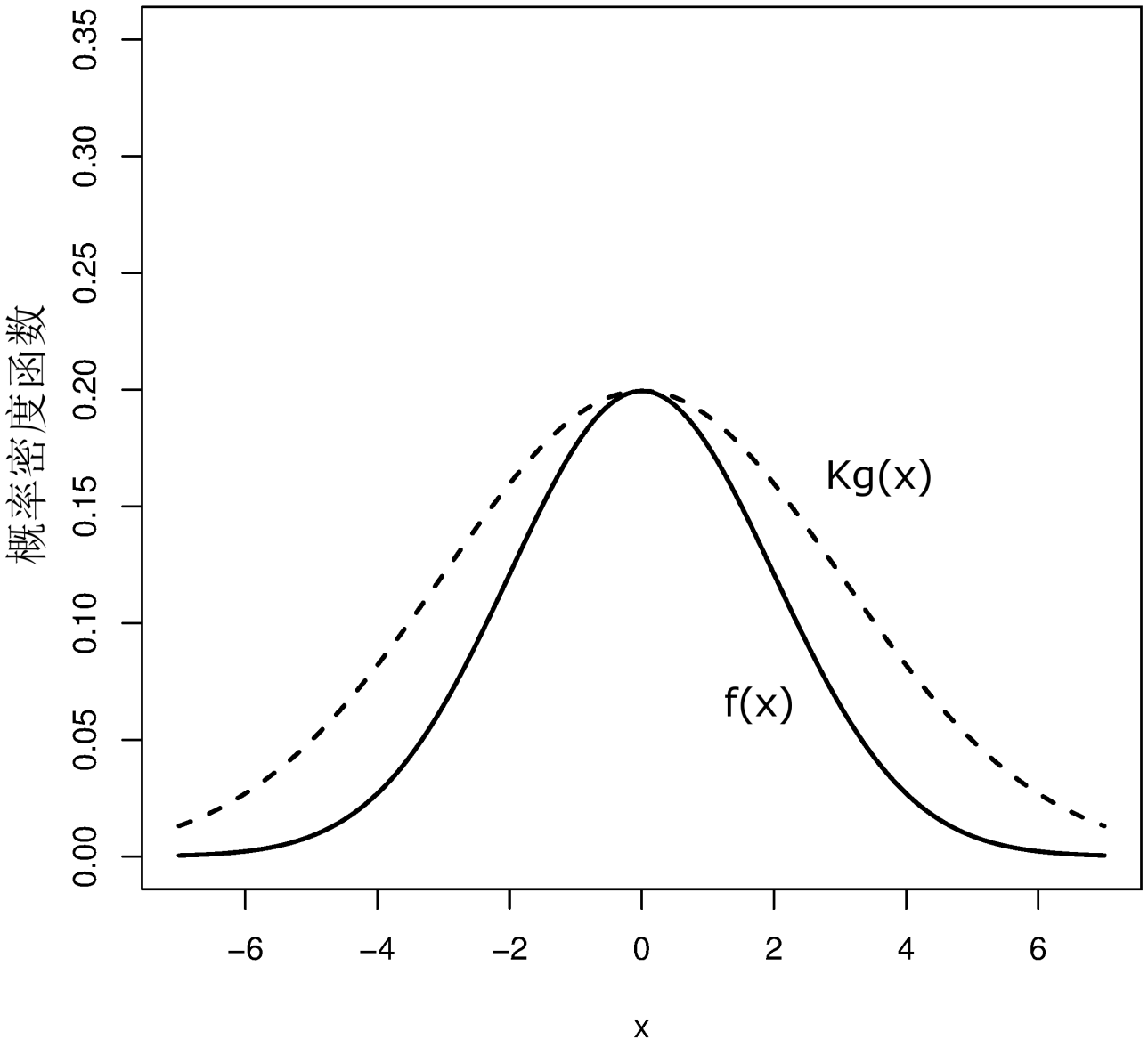


Figure 2: Illustration of rejection sampling with envelop function $Kg(x)$, $K > 1$.

Rejection Sampling

- The rejection sampling algorithm proceeds as follows:
  1. Sample x from the density proportional to $g(x)$.
  2. With probability $\frac{f(x)}{Cg(x)}$, *accept* x as a draw from $f(x)$.
  3. Repeat steps 1 and 2 until enough samples are obtained.

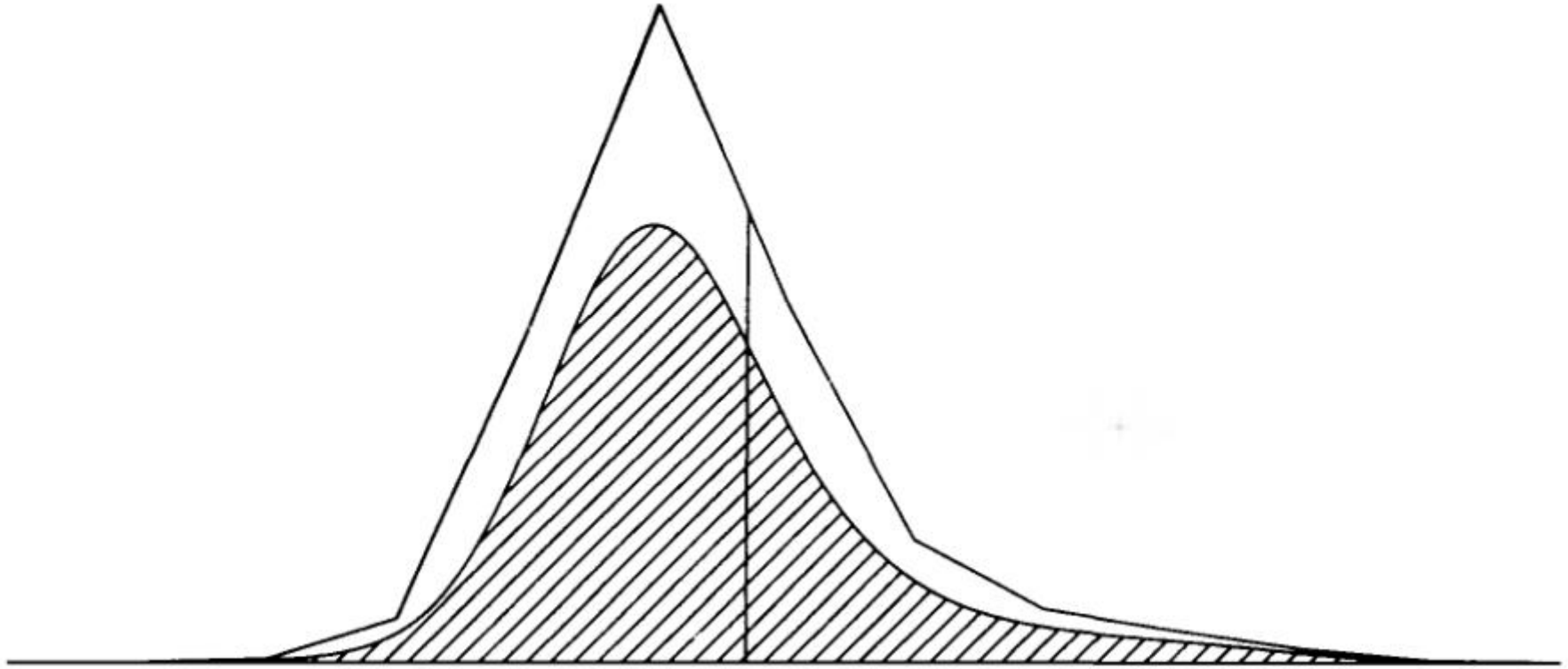


Figure 3: Illustration of rejection sampling. The top curve is an approximation function, $Cg(x)$, and the bottom curve is the target density, $f(x)$. The vertical line indicates a single random draw x from the density proportional to g. The probability that a sampled draw x is accepted is the ratio of the height of the lower curve to the height of the higher curve at the value x.

Rejection Sampling

- A more practical way to implement this is as follows:
  1. Simulate $u \sim \text{Unif}[0, 1]$ and $x \sim g(x)$.
  2. Accept x if $u \leq \frac{f(x)}{Cg(x)}$.
  3. Repeat steps 1 and 2 until we have enough simulated candidates.

Geometric Distribution

- If the probability of success on each trial is p, then the probability that the kth trial (out of k trials) is the first success is

$$P(Y = k) = (1 - p)^{k-1} p, \quad k = 1, 2, \ldots,$$

  and the mean is $1/p$.

- The number of candidates required to obtain a target is geometric with success probability $1/C$. So, $p = 1/C$ here. We hope $C > 1$ to be small.

Justification of Rejection Sampling

- Success probability

$$
\begin{aligned}
& P(x \text{ is accepted}) = P\left(u \le \frac{f(x)}{Cg(x)}\right) \\
= \quad & E_u[I(u \le \frac{f(x)}{Cg(x)})] = E_g\{E_u[I(u \le \frac{f(x)}{Cg(x)})|X = x]\} \\
= \quad & E_g\left[P\left\{u \le \frac{f(x)}{Cg(x)}\middle| X = x\right\}\right] \\
= \quad & E_g[\frac{f(x)}{Cg(x)}] = \int \frac{f(x)}{Cg(x)} g(x)dx = \frac{1}{C}
\end{aligned}
$$

- The distribution of accepted values is $F = \int_{-\infty}^{x} f(t)dt$,

$$P(X \leq x | x \text{ is accepted}) = \frac{P(X \leq x, x \text{ is accepted})}{P(x \text{ is accepted})}$$

$$\begin{aligned}
&= C \times P\left[X \leq x \text{ and } u \leq \frac{f(x)}{Cg(x)}\right] \\
&= C \int_{-\infty}^{x} \left(\int_{0}^{\frac{f(x)}{Cg(x)}} 1du \right) g(x)dx \\
&= \cancel{C} \int_{-\infty}^{x} \frac{f(x)}{\cancel{C}\cancel{g(x)}} \cancel{g(x)}dx \\
&= P_f(X \leq x) = F(x)
\end{aligned}$$

Remarks on Rejection Sampling

- Any constant, $C' > C$, will work. Because the acceptance rate is $\frac{1}{C'}$, C' should be as close to the optimal C as possible.
- Every operation can be performed on the log scale

$$\log u \leq \log f(x) - \log g(x) - \log C.$$

- The higher the dimension of x, the worse rejection sampling is; MCMC may be an option in this case.
- The idea of adaptively estimating C as you go often works well.
- Whether $C = \infty$ or not usually depends on the tails of the candidate, which must be heavier than those of the target.

Posterior

- A common situation is to simulate from $\pi(\boldsymbol{\theta}|\boldsymbol{y}) \propto f(\boldsymbol{y}|\boldsymbol{\theta})\pi(\boldsymbol{\theta})$
- One option is to use π as the candidate,

$$C^* = \sup_{\boldsymbol{\theta}\in\Theta} \frac{\pi(\boldsymbol{\theta}|\boldsymbol{y})}{\pi(\boldsymbol{\theta})} \propto \sup_{\boldsymbol{\theta}\in\Theta} f(\boldsymbol{y}|\boldsymbol{\theta}) = f(\boldsymbol{y}|\hat{\boldsymbol{\theta}}).$$

- Implement the rejection sampling as follows:
  - ✓ Simulate $u \sim \text{Unif}[0,1]$
  - ✓ Simulate $\boldsymbol{\theta} \sim \pi(\boldsymbol{\theta})$
  - ✓ Accept $\boldsymbol{\theta}$ if $u \leq \frac{\pi(\boldsymbol{\theta}|\boldsymbol{y})}{C^*\pi(\boldsymbol{\theta})} = \frac{f(\boldsymbol{y}|\boldsymbol{\theta})\cancel{\pi(\boldsymbol{\theta})}}{C^*\cancel{\pi(\boldsymbol{\theta})}} = \frac{f(\boldsymbol{y}|\boldsymbol{\theta})}{f(\boldsymbol{y}|\hat{\boldsymbol{\theta}})}$

Example

- Suppose $y_1, \ldots, y_n \sim N(\mu, \psi)$. Let $\boldsymbol{\theta} = (\mu, \psi)$. We define the priors: $\mu \sim N(0, \sigma^2)$ and $\psi \sim \mathrm{IG}(\alpha, \beta)$.
- Note that $\hat{\mu} = \bar{y}$ and $\hat{\psi} = \sum_{i=1}^{n}(y_i - \bar{y})^2/n$.
- So, the optimal value is

$$C^* = f(\boldsymbol{y}|\hat{\mu}, \hat{\psi}) = \left(\frac{\sum_{i=1}^{n}(y_i - \bar{y})^2}{n} \right)^{-n/2} e^{-\frac{n}{2}}.$$

- We simulate $u \sim \mathrm{Unif}[0, 1]$, $\mu \sim N(0, \sigma^2)$, $\frac{1}{\psi} \sim$ Gamma(α, β).

- We accept (μ, ψ) if

$$u \leq \frac{f(\boldsymbol{y}|\mu,\psi)}{f(\boldsymbol{y}|\hat{\mu},\hat{\psi})} = \left(\frac{\sum_{i=1}^{n}(y_i-\bar{y})^2}{n\psi}\right)^{\frac{n}{2}} \exp\left\{-\frac{\sum_{i=1}^{n}(y_i-\mu)^2}{2\psi}+\frac{n}{2}\right\}.$$

Posterior Samples

- We repeat this until we have a large enough sample.
  - ✓ Use the upper and lower $\frac{\alpha}{2}$th sample percentiles to estimate the $(1-\alpha)$th posterior equal-tailed credible interval, and the HPD interval.
  - ✓ Posterior mean, variance, median, quantiles
  - ✓ Histogram, density, etc.

Importance Sampling

- Importance sampling is another way to compute expectation $E_f[h(x)]$ using a random sample drawn from an approximation to the target distribution.
- Suppose we can NOT generate random samples from density function $f(x)$, thus we cannot compute $E_f[h(x)]$ using the Monte Carlo method.
- But $g(x)$ is a density function that is easy to generate samples.

- The *non-ratio version* of importance sampling is

$$E_f[h(x)] = \int f(x)h(x)dx = \int \frac{f(x)}{g(x)} g(x)h(x)dx.$$

- So,

$$E_g\left[\frac{f(x)}{g(x)}h(x)\right] \approx \frac{1}{n}\sum_{i=1}^{n} \frac{f(x_i)}{g(x_i)}h(x_i) = \frac{1}{n}\sum_{i=1}^{n} w_i h(x_i),$$

where $x_1, \ldots, x_n$ are samples from $g(x)$, $w_i = \frac{f(x_i)}{g(x_i)}$ is called the *importance ratio* or *importance weight*.

- Note that

$$1 = E_f[1] = E_g\left[\frac{f(x)}{g(x)}\right] \approx \frac{1}{n}\sum_{i=1}^{n} w_i.$$

- The *ratio version* of importance sampling is

$$E_f[h(x)] = \frac{E_f[h(x)]}{E_f[1]} \approx \frac{\sum_{i=1}^{n} w_i h(x_i)}{\sum_{i=1}^{n} w_i}.$$

Remarks

- Initially, the importance sampling estimator (non-ratio version) was proposed to reduce the Monte Carlo standard error of an estimator, i.e., g was chosen to minimize the variance of $\sum_{i=1}^{n} w_i h(x_i)/n$.
- In fact, the minimizer is $g \propto |h(x)|f(x)$; this rule does not hold for the ratio estimator.

Ex: Importance Sampling for Sensitivity Analysis

- We have i.i.d. samples $y_i \sim f(y|\theta)$, for $i = 1, \ldots, n$, and prior $\pi(\theta|\psi_0)$.
- Suppose we have worked really hard to obtain a sample $\{\theta_i\}_{i=1}^n$ from $\pi(\theta|\boldsymbol{y}, \psi_0)$ and we aim to calculate $E[h(\theta)|\boldsymbol{y}, \psi]$ for values of the hyperparameter different from ψ_0.
- Do we have to re-simulate? No, we can just *importance reweigh* the samples we already have, i.e., $f = \pi(\theta|\boldsymbol{y}, \psi)$ and $g = \pi(\theta|\boldsymbol{y}, \psi_0)$. Then with cancelation of the likelihood,

$$E_f[h(\theta)|\boldsymbol{y}, \psi] \approx \frac{\sum_{i=1}^n h(\theta_i) \frac{\pi(\theta_i|\boldsymbol{y},\psi)}{\pi(\theta_i|\boldsymbol{y},\psi_0)}}{\sum_{i=1}^n \frac{\pi(\theta_i|\boldsymbol{y},\psi)}{\pi(\theta_i|\boldsymbol{y},\psi_0)}} = \frac{\sum_{i=1}^n h(\theta_i) \frac{\pi(\theta_i|\psi)}{\pi(\theta_i|\psi_0)}}{\sum_{i=1}^n \frac{\pi(\theta_i|\psi)}{\pi(\theta_i|\psi_0)}}$$

Example: Normal Tail Probabilities

- For $a = 3.5, 4.5, 5.5$, calculate $P(Z > a) = \int_a^\infty \phi(x)dx$, where $Z \sim N(0,1)$.
- "*Naive" approach*: $x_i \sim N(0,1)$, $\int_a^\infty \phi(x)dx = E[I(x > a)]$. So

$$\frac{1}{n}\sum_{i=1}^{n} I(x_i > a) \to \int_a^\infty \phi(x)dx.$$

- *Importance sampling*: $x_i \sim g(x) = ae^{-a(x-a)}$, where $x > a$. Then $\int_a^\infty \phi(x)dx = \int_a^\infty \left[\frac{\phi(x)}{g(x)}\right] g(x)dx$. So,

$$\frac{1}{n}\sum_{i=1}^{n} \frac{\phi(x_i)}{ae^{-a(x_i-a)}} \to \int_a^\infty \phi(x)dx.$$

- *Transform to Uniform* in a finite interval by $y = 1/x$,

$$\int_a^\infty \phi(x)dx = \int_0^{1/a} \frac{\phi(1/y)}{y^2} dy, \quad \text{let } y = 1/x.$$

For $u_i \sim \text{Unif}(0, 1/a)$ with density $g(x) = a$,

$$\frac{1}{n} \sum_{i=1}^n \frac{\phi(1/u_i)}{au_i^2} \to \int_a^\infty \phi(x)dx.$$

Sampling/Importance Resampling

- Finding a good or optimal envelope constant C is a vital step for the use of the rejection method.
- In general, this is not an easy task, especially for high-dimensional cases.
- Another criticism on the rejection method is that it generates 'useless' samples when rejecting the candidate samples.
- As a **non-iterative** sampling procedure, the *sampling/importance resampling* (SIR) method proposed by Rubin (1987, 1988) can bypass the two problems.

SIR without Replacement

- The SIR method generates an approximate i.i.d. sample of size m from the **target density** $f(x)$.
- It consists of two steps: a sampling step and an importance resampling step.
- Specifically, it starts by simulating J i.i.d. samples $\{X^{(j)}\}_{j=1}^{J}$ from an **importance sampling density** or **proposal density** $g(x)$ with the same support $\mathcal{S}_X$ and constructing weights

$$w(X^{(j)}) = f(X^{(j)})/g(X^{(j)})$$

and probabilities

$$\omega_j = \frac{w\big(X^{(j)}\big)}{\sum_{j'=1}^{J} w\big(X^{(j')}\big)}, \quad j = 1, \ldots, J.$$

- Then, a second sample of size $m\ (m \leq J)$ is drawn from the discrete distribution on $\{X^{(j)}\}$ with probabilities $\{\omega_j\}$.
- SIR WITHOUT REPLACEMENT:
  1. Generate $X^{(1)}, \ldots, X^{(J)} \overset{\text{iid}}{\sim} g(\cdot)$;
  2. Select a subset $\{X^{(k_i)}\}_{i=1}^{m}$ from $\{X^{(j)}\}_{j=1}^{J}$ via resampling **without replacement** from the discrete distribution on $\{X^{(j)}\}$ with probabilities $\{\omega_j\}$.

- As expected, the output of the SIR algorithm is good if g is close to f or if J/m is large. A good choice of J should depend on how close g is to f.
- If $g = f$, we can set $J = m$. The poorer is g as an approximation of f, the larger is J compared to m.
- Rubin (1988) showed that the SIR algorithm is exact when $J/m \to \infty$.
- In practice, Rubin (1987) suggested $J/m = 20$ and Smith & Gelfand (1992) recommended $J/m \geq 10$ in their examples.

Example

- Simulate from a density defined in the unit interval.
- Let r be a known positive integer and $X \sim f(x)$, where

$$f(x) = \frac{\pi \sin^r(\pi x)}{B(\frac{1}{2}, \frac{r+1}{2})}, \quad 0 < x < 1. \tag{2}$$

- For example, when $r = 6$, we consider a skewed beta density, say $\text{Beta}(x|2, 4)$, as the importance sampling density $g(x)$, see Figure 1.5(a).
- Thus, the importance weight $w(x) = f(x)/\text{Beta}(x|2, 4)$. We run the SIR algorithm by setting $J =$ 200,000 and $m =$ 20,000.

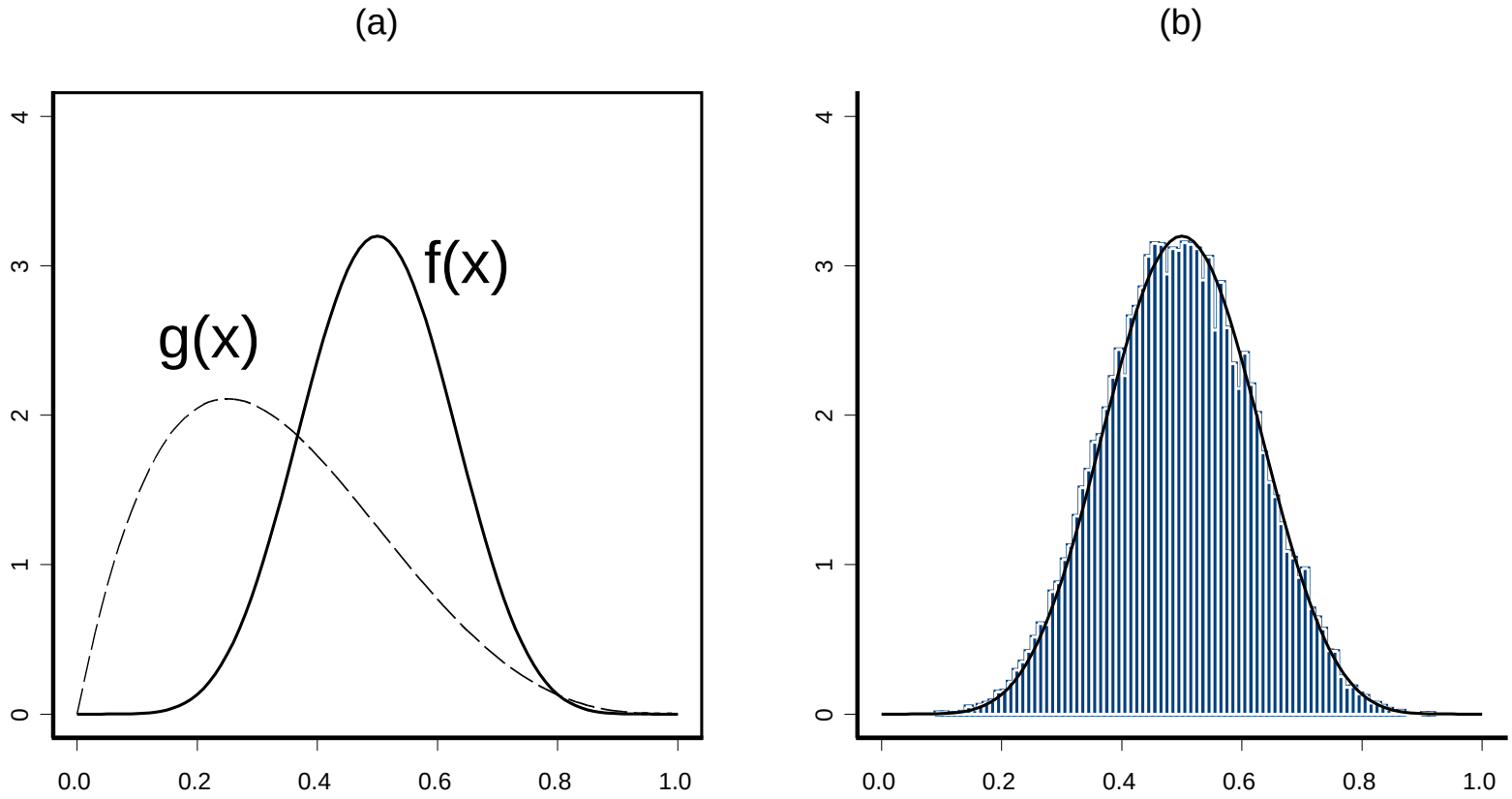


Figure 1.5 The SIR method. (a) The target density $f(x)$ is defined by (2) with $r = 6$ and the importance sampling density $g(x) = \text{Beta}(x|2, 4)$; (b) The histogram of $f(x)$ is obtained by using the SIR method with $J =$ 200,000 and $m =$ 20,000.

Theoretical Justification

- To verify that the SIR method generates samples having approximate density f, let X^* denote a draw from the discrete distribution over $\{X^{(j)}\}$ placing mass ω_j on $X^{(j)}$.
- The cdf of X^* is

$$\Pr(X^* \leq x^*) = \sum_{j=1}^{J} \omega_j I_{(X^{(j)} \leq x^*)} = \frac{\sum_{j=1}^{J} w(X^{(j)}) I_{(X^{(j)} \leq x^*)}}{\sum_{j=1}^{J} w(X^{(j)})}.$$

- When $J \to \infty$, a large quantity of $X^{(j)}$ are available with frequencies according to g and sums become integrals weighted by g, i.e.,

$$\Pr(X^* \leq x^*) \to \frac{\int w(x) I_{(x \leq x^*)} g(x)\, dx}{\int w(x) g(x)\, dx} = \int_{-\infty}^{x^*} f(x)\, dx,$$

  which is the CDF associated with the density f.

SIR with Replacement

- If g is a poor approximation to f, resampling **with replacement** is preferable (Rubin, 1988). Li (2007) stated the SIR algorithm with replacement as follows.

- THE SIR WITH REPLACEMENT:

  1. Generate $X^{(1)}, \ldots, X^{(J)} \overset{\text{iid}}{\sim} g(\cdot)$;
  2. Draw m values $\{Y^{(i)}\}_{i=1}^{m}$ from $X^{(1)}, \ldots, X^{(J)}$ in such a way that $E(q_j | X^{(1)}, \ldots, X^{(J)}) = m\omega_j$, where q_j is the number of copies of $X^{(j)}$ in $\{Y^{(i)}\}_{i=1}^{m}$.

- Smith and Gelfand (1992) used simple weighted random sampling **with replacement** in Step 2 and called their method the **weighted bootstrap**.